

Fedosov manifolds with parallel symplectic Weyl curvature

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ABSTRACT. A classical 1977 result by Roter states that a Riemannian manifold of dimension at least four with parallel Weyl curvature must be locally symmetric or conformally flat. We establish Roter’s theorem in the symplectic category: namely, a Fedosov manifold with parallel symplectic Weyl curvature must be locally symmetric or Weyl-flat.

1. Introduction

By a *Fedosov manifold* we mean, as usual, a triple (M, ω, ∇) where (M, ω) is a connected symplectic manifold and ∇ is a torsionfree connection on M for which $\nabla\omega = 0$ [17]; the connection is explicitly taken to be part of such geometric data because, unlike in the pseudo-Riemannian setting, it is not uniquely determined by ω . Such connections, also called *symplectic*, have played a central role in Fedosov’s seminal work on formal deformation quantization of symplectic manifolds [15] and received considerable attention since then.

The space of all four-times covariant tensors on a symplectic vector space (V, Ω) formally satisfying the symmetries of the curvature tensor R of a symplectic connection admits a natural direct-sum decomposition into $\mathrm{Sp}(V, \Omega)$ -irreducible factors, cf. [22]—some details, needed for later, are recalled in Section 2.1. This means that, for a Fedosov manifold as above, one may write $R = E + W$, where the *symplectic Einstein tensor* E is algebraically built from ω and the Ricci tensor of ∇ , while the *symplectic Weyl tensor* W is fully traceless.

The zeroth-order conditions $E = 0$ (equivalent to $\mathrm{Ric} = 0$) and $W = 0$ have been studied and are rather well-understood. There are many examples of Ricci-flat manifolds which are not Weyl-flat [6, Theorem 4.1], as well as examples of Weyl-flat ones which are not Ricci-flat [1, Corollary 9]. The symplectic Weyl tensor is

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the only obstruction to integrability of a certain almost-complex structure built from ∇ in the bundle of such ω -compatible structures on M [2, Theorem 6.5], while local models for real-analytic Weyl-flat manifolds are also known [9, Theorem 4.3]. The first-order condition $\nabla \text{Ric} = 0$ has also been investigated, and the local product structure of such manifolds has been determined, cf. [8, Proposition 10] and [3, Section 3].

The remaining first-order condition, $\nabla W = 0$, however, does not seem to have received any meaningful attention. This is not surprising from a Riemannian perspective: by a 1977 theorem due to Roter, a Riemannian manifold of dimension at least four having parallel Weyl curvature must be locally symmetric or conformally flat [10, Theorem 2]. Thus, one may naturally wonder if Roter's theorem holds for Fedosov manifolds:

$$(1.1) \quad \begin{array}{l} \text{If a Fedosov manifold has parallel symplectic Weyl} \\ \text{tensor, must it be locally symmetric or Weyl-flat?} \end{array}$$

While (1.1) trivially holds for surfaces, with $W = 0$ for dimensional reasons, its answer in general is not obvious.

On one hand, Roter's original proof via classical tensor calculus methods uses positive-definiteness of the metric in a crucial manner to ultimately conclude from $\|\nabla \text{Ric}\|^2 = 0$ that $\nabla \text{Ric} = 0$, and the result is outright false for indefinite metrics. The so-called *essentially conformally symmetric (ECS)* manifolds—that is, those with $\nabla W = 0$ without $\nabla R = 0$ or $W = 0$ —have been largely investigated by Roter, Derdzinski, and the second author; see for instance [11–14], or [21] for a survey of more recent results. Without any positivity properties assumed for ω , this suggests that the answer to (1.1) might be negative.

On the other hand, from further results by Cahen–Gutt–Rawnsley, it turns out the answer to (1.1) is in fact positive when ∇ is the Levi-Civita connection of a pseudo-Kähler metric having ω as its Kähler form, even when such pseudo-Kähler metric has indefinite signature. Indeed, recalling that a symplectic connection is called *preferred* when it satisfies the gauge condition

$$(1.2) \quad (\nabla_X \text{Ric})(Y, Z) + (\nabla_Y \text{Ric})(Z, X) + (\nabla_Z \text{Ric})(X, Y) = 0$$

for all vector fields X, Y, Z , cf. [5, Section 3], we have that ∇ is preferred whenever $\nabla W = 0$ [8, Remark 1], while in the pseudo-Kähler case ∇ is preferred if and only if $\nabla \text{Ric} = 0$ [8, Proposition 2].

So, which is it? We settle the debate: the answer to (1.1) is *always* positive.

THEOREM A. *A Fedosov manifold with parallel symplectic Weyl tensor must necessarily be locally symmetric or Weyl-flat.*

In other words, there are no “symplectic-ECS manifolds.” The proof strategy we use to establish Theorem A differs from the one adopted by Roter, and instead draws from the representation theory of the real symplectic Lie algebra.

Here is a brief outline of the argument. First, the differential Bianchi identity combined with the condition $\nabla W = 0$ yields the existence of a vector field U on M such that $\nabla_Z \text{Ric}$ is always proportional to $\underline{U} \odot \underline{Z}$ (Proposition 3.2). Then, further differentiating the condition that $R(X, Y) \cdot W = 0$ for all X, Y allows us to identify a certain subspace $\mathcal{S}$ of $\mathfrak{sp}_{2n}(\mathbb{R})$ such that $\mathcal{S} \cdot W = 0$, cf. (2.14). Finally, when $U \neq 0$, we use that $\mathfrak{sp}_{2n}(\mathbb{R})$ is simple and show that the subalgebra of $\mathfrak{sp}_{2n}(\mathbb{R})$ generated by $\mathcal{S}$ is in fact all of $\mathfrak{sp}_{2n}(\mathbb{R})$ (Proposition 2.4). The conclusion follows as then $\mathfrak{sp}_{2n}(\mathbb{R}) \cdot W = 0$ can only happen when $W = 0$ (Proposition 2.2).

The method above can conceivably be used to give an alternative proof of Roter’s original theorem, except perhaps in dimension four as $\mathfrak{so}(4)$ is not simple, and also illustrates its failure for indefinite metrics—see Remarks 2.3 and 2.5 for how the subtler parts of the argument work in the Riemannian setting.

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2. Symplectic curvature tensors and Hamiltonian derivations

Throughout this section, let (V, Ω) be a $2n$ -dimensional symplectic vector space. For every vector $x \in V$, we let $\underline{x} = \Omega(x, \cdot)$ be its image under the isomorphism $V \cong V^*$ induced by Ω .

2.1. The invariant decomposition. By a *symplectic curvature tensor* on V we mean a four-times covariant tensor K on V satisfying

$$(2.1) \quad \begin{aligned} & \text{(i)} \quad K(y, x, z, w) = -K(x, y, z, w), \\ & \text{(ii)} \quad K(x, y, w, z) = K(x, y, z, w), \\ & \text{(iii)} \quad K(x, y, z, w) + K(y, z, x, w) + K(z, x, y, w) = 0, \end{aligned}$$

for all $x, y, z, w \in V$, and we let $\mathcal{R}(V, \Omega)$ denote the vector space of all such symplectic curvature tensors. The *symplectic Kulkarni-Nomizu product* of $F \in \wedge^2 V^*$ and $G \in S^2 V^*$ is defined by

$$(2.2) \quad \begin{aligned} (F \otimes G)(x, y, z, w) &= G(y, z)F(x, w) - G(x, z)F(y, w) \\ &\quad + F(x, z)G(y, w) - F(y, z)G(x, w) + 2F(x, y)G(z, w), \end{aligned}$$

and is easily verified to satisfy (2.1). The *Ricci contraction* of $K \in \mathcal{R}(V, \Omega)$ is defined as $\text{Ric}(K)(y, z) = -\text{tr}_\Omega((x, w) \mapsto K(x, y, z, w))$, where $\text{tr}_\Omega: [V^*]^{\otimes 2} \rightarrow \mathbb{R}$ is described with the aid of a fixed basis for V as $\text{tr}_\Omega(B) = \Omega^{ij} B_{ij}$; here, $[\Omega^{ij}]$ is the inverse of a matrix representation $[\Omega_{ij}]$ of Ω , with the convention that $\Omega^{ij} \Omega_{jk} = \delta_k^i$. A routine computation using (2.2) yields $\text{Ric}(\Omega \otimes G) = 2(n+1)G$, implying that

$$(2.3) \quad \text{the Weyl component of } K, \text{ given by } W(K) = K - \frac{\Omega \otimes \text{Ric}(K)}{2(n+1)},$$

has vanishing Ricci contraction, and leading to the $\text{Sp}(V, \Omega)$ -invariant direct-sum decomposition $\mathcal{R}(V, \Omega) = \text{Im}(\Omega \otimes \cdot) \oplus \text{Ker}(\text{Ric})$.

2.2. Derivations and basic properties. For each integer $k \geq 1$, there is a natural representation of $\mathfrak{sp}(V, \Omega)$ on the space of covariant k -tensors on V , via derivations: namely, for any $A \in \mathfrak{sp}(V, \Omega)$ and $\Theta \in [V^*]^{\otimes k}$, we consider the tensor $A \cdot \Theta \in [V^*]^{\otimes k}$ defined by

$$(2.4) \quad (A \cdot \Theta)(x_1, \dots, x_k) = - \sum_{i=1}^k \Theta(x_1, \dots, Ax_i, \dots, x_k),$$

for all $x_1, \dots, x_k \in V$. Some basic properties of this operation are

$$(2.5) \quad \begin{aligned} & \text{(i) } A \cdot \Omega = 0, \\ & \text{(ii) } A \cdot (B \cdot \Theta) - B \cdot (A \cdot \Theta) = [A, B] \cdot \Theta, \\ & \text{(iii) } A \cdot (F \otimes G) = (A \cdot F) \otimes G + F \otimes (A \cdot G), \end{aligned}$$

for all $A, B \in \mathfrak{sp}(V, \Omega)$, $\Theta \in [V^*]^{\otimes k}$, $F \in \wedge^2 V^*$ and $G \in S^2 V^*$.

Given $K \in [V^*]^{\otimes 4}$ and $x, y \in V$, we consider the endomorphism $\bar{K}(x, y)$ of V characterized by requiring that $\Omega(\bar{K}(x, y)z, w) = K(x, y, z, w)$ for all $z, w \in V$. Note that $\bar{K}(x, y) \in \mathfrak{sp}(V, \Omega)$ whenever K satisfies (2.1-ii). The following auxiliary lemma will be used in Section 2.3.

LEMMA 2.1. *Whenever $A \in \mathfrak{sp}(V, \Omega)$ and $K \in [V^*]^{\otimes 4}$, it holds that*

$$(2.6) \quad [A, \bar{K}(x, y)] = \overline{A \cdot K}(x, y) + \bar{K}(Ax, y) + \bar{K}(x, Ay)$$

for all $x, y \in V$.

PROOF. The relation

$$(2.7) \quad \begin{aligned} (A \cdot K)(x, y, z, w) &= -K(Ax, y, z, w) - K(x, Ay, z, w) \\ &\quad - K(x, y, Az, w) - K(x, y, z, Aw), \end{aligned}$$

immediate from (2.4), may be rewritten as

$$(2.8) \quad \begin{aligned} \Omega(\overline{A \cdot K}(x, y)z, w) &= -\Omega(\bar{K}(Ax, y)z, w) - \Omega(\bar{K}(x, Ay)z, w) \\ &\quad + \Omega([A, \bar{K}(x, y)]z, w) \end{aligned}$$

by using that $A \in \mathfrak{sp}(V, \Omega)$ to combine the last two terms in the right side of (2.7) into the term involving $[A, \bar{K}(x, y)]$. Eliminating z, w from (2.8) and rearranging it, we obtain (2.6). $\square$

Finally, we show that the representation (2.4) restricted to $\mathcal{R}(V, \Omega)$ has no nontrivial invariants.

PROPOSITION 2.2. *If $K \in \mathcal{R}(V, \Omega)$ has $A \cdot K = 0$ for every $A \in \mathfrak{sp}(V, \Omega)$, then necessarily $K = 0$.*

PROOF. Due to (2.1-ii), it suffices to show that $K(x, y, z, z) = 0$ for all vectors $x, y, z \in V$. We achieve this by establishing that

$$(2.9) \quad (i) \ K(x, y, z, z) - 2K(y, z, x, z) = 0, \quad (ii) \ K(x, y, z, z) + 2K(y, z, x, z) = 0,$$

instead. With $\underline{x} \otimes x \in \mathfrak{sp}(V, \Omega)$, we may use (2.4) together with (2.1-i) to directly compute that

$$(2.10) \quad 0 = -((\underline{x} \otimes x) \cdot K)(x, y, y, y) = 2\Omega(x, y)K(x, y, x, y).$$

As the set $\{(x, y) \in V \times V : \Omega(x, y) \neq 0\}$ is dense in $V \times V$ due to nondegeneracy of Ω , continuity of K leads via (2.10) to $K(x, y, x, y) = 0$ for all $x, y \in V$. Polarizing this last relation in the variable x we obtain

$$(2.11) \quad K(x, y, z, y) + K(z, y, x, y) = 0, \quad \text{for all } x, y, z \in V,$$

and further polarizing (2.11) in the variable y yields

$$(2.12) \quad K(x, y, z, z) + K(x, z, z, y) + K(z, y, x, z) + K(z, z, x, y) = 0.$$

Applying (2.1-i) to the third and fourth terms, (2.1-ii) to the second term, and then (2.11) with the roles of y and z switched, (2.12) reduces to (2.9-i). Obtaining (2.9-ii) as well is now straightforward: apply (2.1-i) and then (2.11) with the roles of y and z switched to the third term of (2.1-iii) with $w = z$. $\square$

REMARK 2.3. Proposition 2.2 does not hold as stated in the pseudo-Riemannian setting. Namely, whenever (V, g) is a pseudo-Euclidean vector space and a curvaturelike tensor R satisfies that $A \cdot R = 0$ for all $A \in \mathfrak{so}(V, g)$, then R must be a multiple of R_0 given by $R_0(x, y, z, w) = g(y, z)g(x, w) - g(x, z)g(y, w)$. Indeed, in this case every $\text{SO}(V, g)$ -invariant 4-tensor R must be a linear combination of $g(x, y)g(z, w)$, $g(x, z)g(y, w)$, and $g(x, w)g(y, z)$, cf. [18, Theorem 5.3.3], and imposing (2.1-i), skew-symmetry in the last two arguments, and (2.1-iii), leads to the claimed conclusion. In particular, this implies that if W is a *Weyl* curvaturelike tensor with $A \cdot W = 0$ for all $A \in \mathfrak{so}(V, g)$, then $W = 0$. It is also part of [18, Theorem 5.3.3] that any $\text{Sp}(V, \Omega)$ -invariant 4-tensor is a linear combination of $\Omega(x, y)\Omega(z, w)$,

$\Omega(x, z)\Omega(y, w)$, and $\Omega(x, w)\Omega(y, z)$, making the real content of Proposition 2.2 that no such nontrivial combinations survive (2.1).

2.3. Subspaces ultimately annihilating W . A substantial part of the proof of Theorem A may be carried out in the general setting of symplectic vector spaces. We thus present it here first, and then invoke it in Section 4 at the right moment. Observe that the mapping

$$(2.13) \quad \mathfrak{sp}(V, \Omega) \ni A \mapsto \Omega(\cdot, A\cdot) \in S^2V^* \text{ is an isomorphism of vector spaces,}$$

being injective due to nondegeneracy of Ω , and between spaces of the same dimension $n(2n+1)$. Then, given a vector $u \in V$, we consider

$$(2.14) \quad \begin{aligned} & \text{the subspace } \mathcal{S} \text{ of } \mathfrak{sp}(V, \Omega) \text{ corresponding under} \\ & (2.13) \text{ to the subspace of } S^2V^* \text{ spanned by elements} \\ & \text{of the form } \Omega \bigcirc (\underline{u} \odot \underline{z})(x, y, \cdot, \cdot), \text{ with } x, y, z \in V. \end{aligned}$$

It is in the next result that simplicity of $\mathfrak{sp}(V, \Omega)$ plays a crucial role. Below, for each subspace L of V , we set $L^\Omega = \{x \in V : \Omega(x, y) = 0 \text{ for all } y \in L\}$.

PROPOSITION 2.4. *If $u \neq 0$ and $2n \geq 4$, the subalgebra $\mathfrak{u}$ generated by the subspace $\mathcal{S}$ in (2.14) in fact equals all of $\mathfrak{sp}(V, \Omega)$.*

PROOF. Our initial claim is that

$$(2.15) \quad \mathcal{S} \not\subseteq \mathfrak{p}, \text{ where } \mathfrak{p} = \{A \in \mathfrak{sp}(V, \Omega) : Au \in \mathbb{R}u\} \text{ is the stabilizer of } \mathbb{R}u.$$

As $A \in \mathfrak{p}$ if and only if $\Omega(Au, w) = 0$ for all $w \in (\mathbb{R}u)^\Omega$, we establish (2.15) by exhibiting $x, y \in V$ and $w \in (\mathbb{R}u)^\Omega$ such that $(\Omega \bigcirc (\underline{u} \odot \underline{x}))(x, y, u, w) \neq 0$. Namely, as $u \neq 0$ we may take $x \notin (\mathbb{R}u)^\Omega$, while the assumption that $2n \geq 4$ allows us to take $y \in (\mathbb{R}u)^\Omega \setminus (\mathbb{R}u)$ and $w \in (\mathbb{R}u)^\Omega$ with $\Omega(y, w) \neq 0$. With such choices, (2.2) immediately leads to $(\Omega \bigcirc (\underline{u} \odot \underline{x}))(x, y, u, w) = \Omega(x, u)^2 \Omega(y, w) \neq 0$, as required.

The next step is to show that

$$(2.16) \quad [\mathfrak{p}, \mathfrak{u}] \subseteq \mathfrak{u}, \text{ so that } \mathfrak{u} + \mathfrak{p} \text{ is a subalgebra of } \mathfrak{sp}(V, \Omega).$$

Let $A \in \mathfrak{p}$ and $B \in \mathcal{S}$ correspond under (2.13) to $(\Omega \bigcirc (\underline{u} \odot \underline{z}))(x, y, \cdot, \cdot)$, for some $x, y, z \in V$; write also $Au = \lambda u$, with $\lambda \in \mathbb{R}$. Combining the fact that $A \cdot \underline{v} = \underline{Av}$ for all $v \in V$, relations (2.5-i) and (2.5-iii), and Lemma 2.1, it follows that the commutator $[A, B]$ corresponds under (2.13) to the sum of the four terms

$$(2.17) \quad \begin{aligned} & \lambda(\Omega \bigcirc (\underline{u} \odot \underline{z}))(x, y, \cdot, \cdot), \quad (\Omega \bigcirc (\underline{u} \odot \underline{Az}))(x, y, \cdot, \cdot), \\ & (\Omega \bigcirc (\underline{u} \odot \underline{z}))(Ax, y, \cdot, \cdot), \quad \text{and} \quad (\Omega \bigcirc (\underline{u} \odot \underline{z}))(x, Ay, \cdot, \cdot), \end{aligned}$$

so that $[A, B] \in \mathcal{S}$ by (2.14), and hence $[\mathfrak{p}, \mathcal{S}] \subseteq \mathcal{S}$. The Jacobi identity guarantees that $\mathfrak{g} = \{B \in \mathfrak{u} : [\mathfrak{p}, B] \subseteq \mathfrak{u}\}$ is a subalgebra of $\mathfrak{u}$, while at the same time the previous argument yields $\mathcal{S} \subseteq \mathfrak{g}$. Thus $\mathfrak{g} = \mathfrak{u}$, establishing (2.16).

Finally, observe that $\mathfrak{p}$ —being the stabilizer of a line—is maximal as a subalgebra of $\mathfrak{sp}(V, \Omega)$ [4, pp. 201–202, (III)]. As $\mathfrak{u} + \mathfrak{p}$ is a subalgebra of $\mathfrak{sp}(V, \Omega)$ by (2.16), and contains $\mathfrak{p}$ strictly by (2.15), we necessarily have that $\mathfrak{u} + \mathfrak{p} = \mathfrak{sp}(V, \Omega)$. This last relation, in turn, immediately implies that $\mathfrak{u}$ is an ideal of $\mathfrak{sp}(V, \Omega)$. With $\mathfrak{u} \neq 0$ and $\mathfrak{sp}(V, \Omega)$ being simple, cf. [19, Theorem 6.105], it follows that $\mathfrak{u} = \mathfrak{sp}(V, \Omega)$ as required. $\square$

REMARK 2.5. Maximality of $\mathfrak{p}$ in (2.15) also holds in the Riemannian setting. Namely, if for a pseudo-Euclidean vector space (V, g) and $v \in V \setminus \{0\}$ we set $\mathfrak{p} = \{A \in \mathfrak{so}(V, g) : Av = 0\}$, then $\mathfrak{p}$ is guaranteed to be maximal as a subalgebra of $\mathfrak{so}(V, g)$ whenever g is positive-definite. Indeed, we first note that the mapping $\varphi: v^\perp \rightarrow \mathfrak{so}(V, g)/\mathfrak{p}$ given by $\varphi(w) = v \wedge w + \mathfrak{p}$, easily seen to be $\mathfrak{p}$ -equivariant, is an isomorphism if and only if $g(v, v) \neq 0$: it has $\text{Ker}(\varphi) = \{w \in v^\perp : g(v, v)w = 0\}$ and $\varphi(g(v, v)^{-1}Av) = A + \mathfrak{p}$ for all $A \in \mathfrak{so}(V, g)$, whenever $g(v, v) \neq 0$. We then claim that $v^\perp$ is irreducible as a $\mathfrak{p}$ -module whenever g is positive-definite, thus implying that $\mathfrak{p}$ is maximal. To establish this, we show that if a nontrivial subspace $Z \subseteq v^\perp$ is $\mathfrak{p}$ -invariant and not null, then $Z = v^\perp$: fix $z \in Z \setminus \{0\}$ such that $g(z, z) \neq 0$ and observe that, as $w = g(z, z)^{-1}(z \wedge w)z \in Z$ whenever $w \in z^\perp \cap v^\perp$ (using that $z \wedge w \in \mathfrak{p}$), we obtain $z^\perp \cap v^\perp \subseteq Z$; the conclusion follows since, decomposing any $x \in v^\perp$ as $x = \lambda z + x_0$ with $\lambda \in \mathbb{R}$ and $x_0 \in z^\perp \cap v^\perp$, both λz and x_0 lie in Z .

3. Differential identities involving the Ricci tensor

Hereafter, let (M, ω, ∇) be a $2n$ -dimensional Fedosov manifold, R its curvature tensor, and $W = W(R)$ be its symplectic Weyl tensor as in (2.3). As in Section 2, for any vector field X on M we let $\underline{X} = \omega(X, \cdot)$ be the corresponding 1-form.

Recall that, in addition to the symmetries (2.1), R also satisfies the differential Bianchi identity $d^\nabla R = 0$, where d^∇ is the covariant exterior derivative operator and R is regarded as a 2-form valued in $\text{End}(TM)$. As ∇ is torsionfree and parallelizes ω , this identity may be rewritten in terms of the fully-covariant curvature tensor—with the convention that $R(X, Y, Z, S) = \omega(R(X, Y)Z, S)$ —as

$$(3.1) \quad (\nabla_X R)(Y, Z, S, T) + (\nabla_Y R)(Z, X, S, T) + (\nabla_Z R)(X, Y, S, T) = 0.$$

Furthermore, the covariant exterior derivative of Ric is given by

$$(3.2) \quad (d^\nabla \text{Ric})(X, Y)Z = (\nabla_X \text{Ric})(Y, Z) - (\nabla_Y \text{Ric})(X, Z),$$

if Ric is regarded as a 1-form valued in T^*M .

In [7, Lemma 1(i)] it was shown that, whenever $W = 0$, there is a vector field U on M such that

$$(3.3) \quad \nabla_Z \text{Ric} = -\frac{1}{2n+1} \underline{U} \odot \underline{Z}, \quad \text{for all } Z \in \mathfrak{X}(M).$$

Up to a constant, U is dual to the divergence 1-form $(\text{div}_\omega \text{Ric})_k = \omega^{ij} \nabla_i R_{jk}$, which plays the role of “differential of the scalar curvature,” cf. [16, Section 3]. A major step towards the proof of Theorem A is to replace $W = 0$ with the weaker condition $\nabla W = 0$. In order to do so, for each $Z \in \mathfrak{X}(M)$ we consider the differential forms

$$(3.4) \quad \begin{aligned} \phi_Z &\in \Omega^1(M) \text{ and } \sigma_Z \in \Omega^2(M), \text{ given by} \\ \phi_Z &= (\nabla \text{Ric})(Z, \cdot) \text{ and } \sigma_Z = (d^\nabla \text{Ric})(\cdot, \cdot)Z, \end{aligned}$$

and explore relations between them.

LEMMA 3.1. *If $\nabla W = 0$, then $\omega \wedge \phi_Z + \underline{Z} \wedge \sigma_Z = 0$ for all $Z \in \mathfrak{X}(M)$.*

PROOF. Under the condition $\nabla W = 0$, as covariant derivatives naturally satisfy a Leibniz rule for the symplectic Kulkarni-Nomizu product, it follows from (2.3) that (3.1) reduces to

$$(3.5) \quad \begin{aligned} &(\omega \odot \nabla_X \text{Ric})(Y, Z, S, T) + (\omega \odot \nabla_Y \text{Ric})(Z, X, S, T) \\ &+ (\omega \odot \nabla_Z \text{Ric})(X, Y, S, T) = 0. \end{aligned}$$

By using (2.2), grouping terms, and recognizing the definition of ϕ_Z in (3.4), we have that

$$(3.6) \quad \begin{aligned} \frac{1}{2}(\omega \odot \nabla_X \text{Ric})(Y, T, Z, Z) &= (\nabla_X \text{Ric})(T, Z)\omega(Y, Z) \\ &- (\nabla_X \text{Ric})(Y, Z)\omega(T, Z) + \omega(Y, T)\phi_Z(X). \end{aligned}$$

We now cyclically sum (3.6) over X, Y, T . The left side of (3.6) sums to zero by (3.5), while the last term in the right side sums to $(\omega \wedge \phi_Z)(X, Y, T)$. The remaining six terms in the cyclic sum coming from the last two terms in (3.6) unaccounted for are grouped in pairs with the aid of (3.2) and, recognizing the definition of σ_Z in (3.4), amount to $(\underline{Z} \wedge \sigma_Z)(X, Y, T)$ as required. $\square$

To conclude this section, consider the Lefschetz and dual-Lefschetz operators $L: \Omega^k(M) \rightarrow \Omega^{k+2}(M)$ and $\Lambda: \Omega^k(M) \rightarrow \Omega^{k-2}(M)$, defined by

$$(3.7) \quad L\alpha = \omega \wedge \alpha \quad \text{and} \quad \Lambda\alpha = \frac{1}{2} \text{tr}_\omega((X, Y) \mapsto \alpha(X, Y, \cdot, \dots, \cdot))$$

for $\alpha \in \Omega^k(M)$; see e.g. [20, Section 3.2]. A straightforward calculation (e.g., using Darboux coordinates) shows that

$$(3.8) \quad \Lambda(\omega \wedge \alpha) = (n-1)\alpha \quad \text{and} \quad \Lambda(\underline{X} \wedge \beta) = (\Lambda\beta)\underline{X} - \beta(X, \cdot)$$

whenever $\alpha \in \Omega^1(M)$ and $\beta \in \Omega^2(M)$.

PROPOSITION 3.2. *If $2n \geq 4$ and $\nabla W = 0$, there is a vector field U on M for which (3.3) holds. In particular, (M, ω, ∇) is locally symmetric if and only if $U = 0$.*

PROOF. The vector field U is defined by requiring that $\omega(U, Z) = -\Lambda\sigma_Z$ for all $Z \in \mathfrak{X}(M)$, where Λ and σ_Z as in (3.7) and (3.4); note that the dependence of $\Lambda\sigma_Z$ on Z is indeed tensorial, making U well-defined. Combining the definitions of ϕ_Z and σ_Z with (3.2), we see that

$$(3.9) \quad \sigma_Z(Z, X) = (\nabla_Z \text{Ric})(X, Z) - \phi_Z(X)$$

for all $X, Z \in \mathfrak{X}(M)$. At the same time, applying Λ to the identity obtained in Lemma 3.1 and using (3.8), we see that $(n-1)\phi_Z + (\Lambda\sigma_Z)\underline{Z} - \sigma_Z(Z, \cdot) = 0$. Substituting (3.9) into this last relation and using the definition of U , it follows that $n(\nabla_X \text{Ric})(Z, Z) = \omega(U, Z)\omega(Z, X) + (\nabla_Z \text{Ric})(X, Z)$. Polarization in the variable Z now yields

$$(3.10) \quad 2n(\nabla_X \text{Ric})(Y, Z) = -(\underline{U} \odot \underline{X})(Y, Z) + (\nabla_Z \text{Ric})(X, Y) + (\nabla_Y \text{Ric})(X, Z).$$

Switching the roles of X and Y in (3.10) and adding the resulting identity to (3.10) itself, we obtain

$$(3.11) \quad (2n-1)(\nabla_X \text{Ric}(Y, Z) + (\nabla_Y \text{Ric})(X, Z)) = 2(\nabla_Z \text{Ric})(X, Y) + (\underline{U} \odot \underline{Z})(X, Y).$$

This time switching the roles of X and Z in (3.10), and substituting it into (3.11), it follows that

$$(3.12) \quad (2n-1)(2n\nabla_Z \text{Ric} + \underline{U} \odot \underline{Z}) = 2\nabla_Z \text{Ric} + \underline{U} \odot \underline{Z},$$

as symmetric tensor fields. Dividing both sides of (3.12) by $2-2n \neq 0$ and solving for $\nabla_Z \text{Ric}$, we finally arrive at (3.3). $\square$

REMARK 3.3. Both Lemma 3.1 and Proposition 3.2 (but not necessarily Theorem A itself) hold under the more general assumption that W satisfies (3.1).

4. Proof of Theorem A

We continue with the setup from the previous section, and assume from here onwards that $2n \geq 4$ and $\nabla W = 0$. The latter implies that $R(X, Y) \cdot W = 0$ for all X, Y , cf. (2.4), and further differentiation leads to $(\nabla_Z R)(X, Y) \cdot W = 0$ for all X, Y, Z . Taking into account (2.3) and Proposition 3.2, we obtain

$$(4.1) \quad (\omega \odot (\underline{U} \odot \underline{Z}))(X, Y) \cdot W = 0, \quad \text{for all } X, Y, Z \in \mathfrak{X}(M).$$

If $\nabla R \neq 0$, and hence $\nabla \text{Ric} \neq 0$, Proposition 3.2 yields a point $p \in M$ such that $U_p \neq 0$. We then set $(V, \Omega) = (T_p M, \omega_p)$ and consider the subspace $\mathcal{S}$ of $\mathfrak{sp}(V, \Omega)$ defined via (2.14) for $u = U_p$. It is clear from (4.1) that $\mathcal{S} \cdot W_p = 0$, so that (2.5-ii) and Proposition 2.4 together lead to $\mathfrak{sp}(V, \Omega) \cdot W_p = 0$. Proposition 2.2 then implies that $W_p = 0$ and, as $\nabla W = 0$ and M is connected, it finally follows that $W = 0$. $\square$

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